- Previously, training would immediately fail with NaNs.
- This was traced to a recursive all-reduce issue in DTensor::reduce_grad.
- Cause - reduce_grad() was modifying the autograd state of the same underlying tensor in-place.
- If multiple operations required a gradient reduction, they would "stack" all-reduces on the same memory, leading to exponential growth.
- I modified reduce_grad() to return a view of the tensor.
- This allows the autograd engine to treat the reduction as a distinct node in the graph without corrupting the original tensor's state.

- After fixing NaNs, the loss sat at ~11.2 indefinitely.
- This was caused by several compounding issues in gradient scaling.

Cross-Entropy Scaling

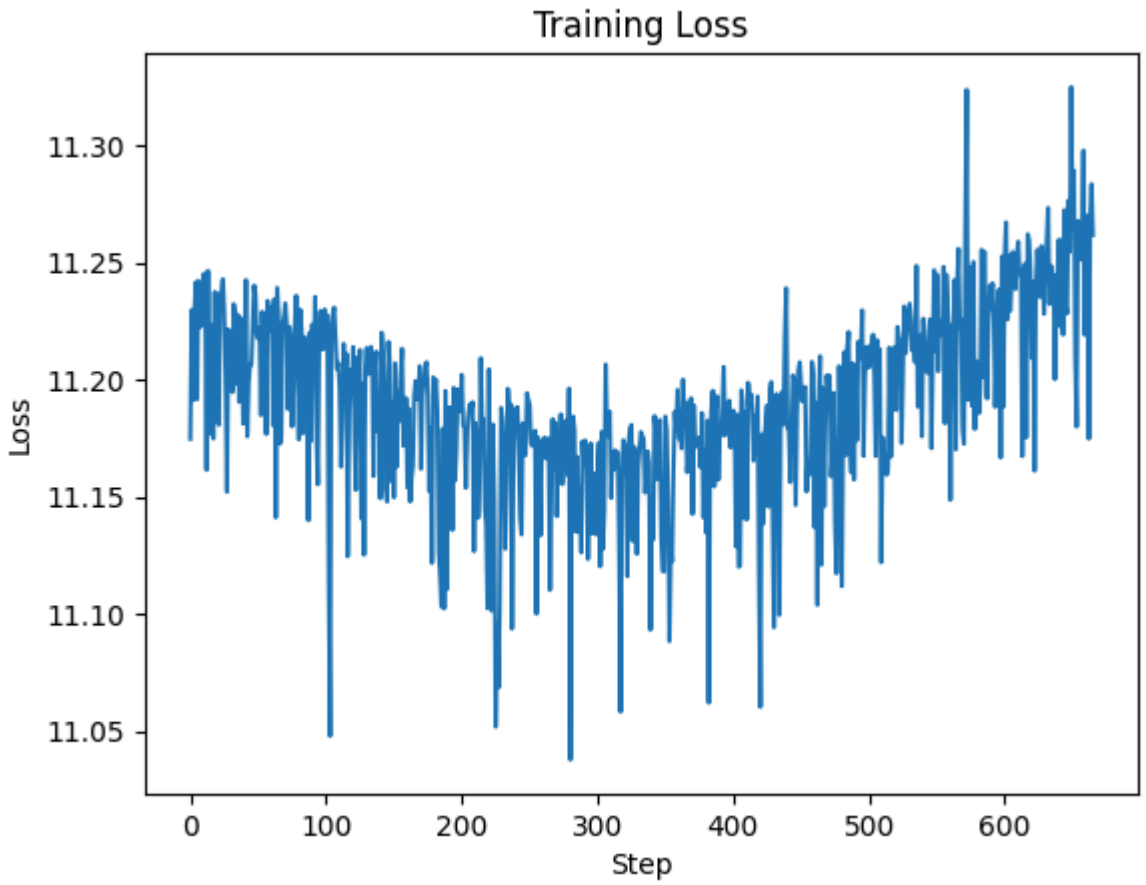
- In distributed_sparse_cross_entropy_loss, the grad_fn was incorrectly attached to the mean_loss.
- The manual gradient calculation produced the gradient for the summed loss.
- By attaching it to mean_loss, the autograd engine applied an additional reduction (where
- This made the gradients nearly zero.
- Attached the grad_fn to sample_losses, ensuring the scaling from reduce_mean is correctly applied to our calculated gradients.

DLinear Gradient Missing All-Reduce

- In Column-Parallel layers, the input is replicated.
- Each GPU computes a partial gradient for that input based on its shard of the weights.
- These must be summed across all GPUs.
- Added input.reduce_grad() in the DLinear::forward pass.

```
[Step 40] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: inf | First bad val at [0]: nan
[Step 40] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 40] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    40 | loss: -inf | lr 4.8810e-05 | norm: nan | dt: 1753.45ms | tok/sec: 37375.44
[Step 41] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 41] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 41] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    41 | loss: nan | lr 5.0000e-05 | norm: nan | dt: 1720.98ms | tok/sec: 38080.53
[Step 42] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 42] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 42] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    42 | loss: nan | lr 5.1190e-05 | norm: nan | dt: 1724.44ms | tok/sec: 38004.27
[Step 43] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 43] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 43] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    43 | loss: nan | lr 5.2381e-05 | norm: nan | dt: 1728.16ms | tok/sec: 37922.43
[Step 44] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 44] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 44] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    44 | loss: nan | lr 5.3571e-05 | norm: nan | dt: 1720.65ms | tok/sec: 38087.89
[Step 45] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 45] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 45] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    45 | loss: nan | lr 5.4762e-05 | norm: nan | dt: 1719.29ms | tok/sec: 38118.12
[Step 46] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 46] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 46] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    46 | loss: nan | lr 5.5952e-05 | norm: nan | dt: 1728.65ms | tok/sec: 37911.58
[Step 47] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 47] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 47] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    47 | loss: nan | lr 5.7143e-05 | norm: nan | dt: 1726.95ms | tok/sec: 37948.88
[Step 48] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 48] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 48] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
step    48 | loss: nan | lr 5.8333e-05 | norm: nan | dt: 1745.13ms | tok/sec: 37553.57
[Step 49] NaN gradient in wte (idx 0) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 49] NaN gradient in layer_1.mlp_w_up (idx 10) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
[Step 49] NaN gradient in layer_1.mlp_b_up (idx 11) | Norm: nan | Min: 10000000150474662198766888855040.00 | Max: -10000000150474662198766888855040.00 | First bad val at [0]: nan
```

Current graph -



Found no NaN loss till 600 steps, the training seems stable.

Although the loss is behaving abnormally, so will look into fixing that.

Regards
Anuj Kumar